Assignment 1

AI-based Village Pond Planning System

Background

Water conservation is a major challenge in rural areas. One effective solution is the construction of ponds at suitable locations to harvest rainwater. Selecting these locations requires analysis of terrain elevation, catchment areas, government land availability, rainfall patterns, and estimated storage capacity.

In this assignment, you will develop a web application that assists village administrators in identifying suitable locations for pond construction using geospatial and satellite data.

Objective

Develop a complete web application capable of analyzing terrain and rainfall information to recommend suitable locations for pond construction. The system should estimate catchment areas, rainfall accumulation, pond depth, and storage capacity, and present the results through an interactive map interface.

Functional Requirements

The application shall support the following features.

1. Display satellite imagery for a selected village.
2. Visualize contour maps.
3. Identify available land suitable for pond excavation.
4. Estimate the catchment area contributing runoff to the selected location.
5. Query historical rainfall data using publicly available APIs.
6. Estimate runoff volume using rainfall and catchment information.
7. Recommend an appropriate pond depth and approximate storage capacity.
8. Overlay all results containing:
    - Selected pond location
    - Catchment area
    - Annual rainfall statistics
    - Estimated runoff volume
    - Recommended pond dimensions
    - Maps and visualizations

Suggested Technology Stack

Students are free to choose their implementation. A suggested stack is

- Python

- Flask or FastAPI
- OpenCV
- MongoDB/PostgreSQL
- Elevation APIs (OpenZenith, etc.) and Rainfall APIs (IMD, Open-Meteo, NASA POWER, etc.)
- A basic frontend libray

Deliverables

- Complete source code
- Installation guide
- API documentation
- An accessible Front-end for users
- Final technical report

Important Dates

- **High-Level Design (HLD) Submission:** August 10
- **Prototype Demonstration and Feedback:** Lab Hours
- **Final Submission and Demonstration:** September 5

Evaluation

Criterion	Marks
System functionality	35
Terrain and catchment analysis	20
Frontend and visualization	5
Software design and code quality	15
System Design and Management	15
Documentation and report	10
Total	**100**

LLM Usage Policy

The use of Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, GitHub Copilot, and similar AI-assisted coding tools is permitted for brainstorming, understanding concepts, debugging, documentation, code refactoring, and improving code quality. However, students are expected to understand and be able to explain every component of their submitted work. Direct submission of AI-generated code without verification, modification, or understanding is considered unacceptable. During demonstrations, students may be asked to explain design decisions, algorithms, implementation details, and justify the use of external libraries or AI-generated code. Failure to satisfactorily explain the submitted work may result in reduction of marks or rejection of the submission. Students are instructed to cite use of AI tools in their project report and maintain academic integrity throughout the assignment, failure to do so will attract disciplinary action.

**Used AI tools for formatting these pages*